

Q2 Find the number of ways a man, woman, boy and girl can be selected from 5 men, 6 women, 2 boys, and 4 girls.

Solution.

By the multiplication principle (MP), we have $5 \times 6 \times 2 \times 4 = 240$ ways to select a man, a woman, a boy, and a girl from 5 men, 6 women, 2 boys, and 4 girls. $\square$

Q14 Determine the number of nonnegative integer solutions to the pair of equations

$$x_1 + x_2 + x_3 = 6, \quad x_1 + x_2 + \cdots + x_5 = 15$$

where $x_i \geq 0$ for all $i = 1, 2, 3, 4, 5$.

Solution.

We have two equations as the following:

$$x_1 + x_2 + x_3 = 6 \tag{1}$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 15 \tag{2}$$

Subtracting Equation 1 from Equation 2, we have:

$$x_1 + x_2 + x_3 = 6 \tag{3}$$

$$x_4 + x_5 = 9 \tag{4}$$

Thus, we have separated the problem into two smaller independent problems; the first from Equation 3, and the second from Equation 4, given the constraint $x_i \geq 0$ for all $i = 1, 2, 3, 4, 5$.

There are $\binom{6+3-1}{6} = \binom{8}{6} = 28$ ordered triples (x_1, x_2, x_3) that satisfy Equation 3 and $x_i \geq 0$ for $i = 1, 2, 3$.

There are also $\binom{9+2-1}{9} = \binom{10}{9} = 10$ ordered pairs (x_4, x_5) that satisfy Equation 4 and $x_i \geq 0$ for $i = 4, 5$.

Therefore, by MP, there are $28 \times 10 = 280$ ordered 5-tuples $(x_1, x_2, x_3, x_4, x_5)$ that satisfy the given pair of equations and all $x_i \geq 0$ for all $i = 1, 2, 3, 4, 5$. $\square$